Supplementary Figures and Tables

Mechanistic learning to predict and understand minimal residual disease in acute lymphoblastic leukemia

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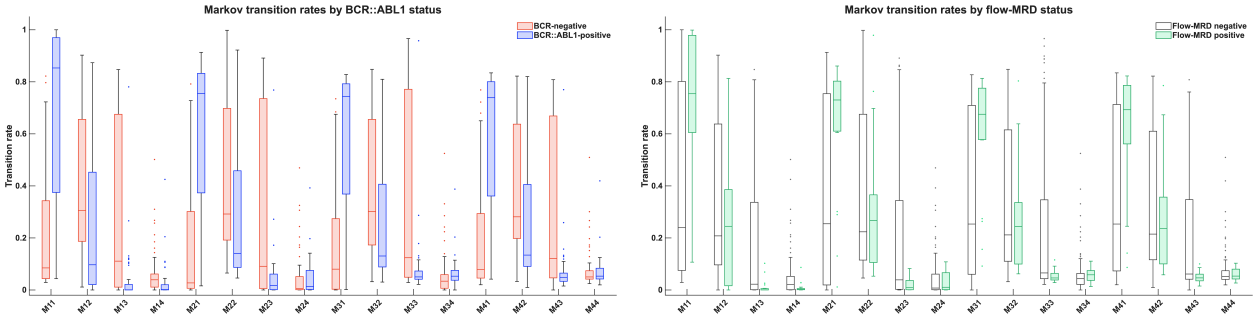


Figure S1: Mean Markov transition parameters by (A) BCR::ABL1 and (B) flow-MRD status. Each specimen's parameters are averaged across 100 stochastic fits. Boxes show medians and interquartile ranges.

Transition	N	Median within-specimen SD	IQR of within-specimen SD	Mean within-specimen SD	Maximum within-specimen SD
M11	104	0.0754	0.0225–0.1119	0.0691	0.1497
M12	104	0.0895	0.0181–0.1353	0.0827	0.1710
M13	104	0.0092	0.0005–0.1228	0.0536	0.1631
M14	104	0.0049	0.0009–0.0695	0.0345	0.1562
M21	104	0.0999	0.0217–0.1336	0.0823	0.1655
M22	104	0.1142	0.0586–0.1350	0.0995	0.1710
M23	104	0.0201	0.0018–0.0925	0.0469	0.1568
M24	104	0.0075	0.0012–0.0646	0.0373	0.1595
M31	104	0.1419	0.0952–0.1504	0.1145	0.1720
M32	104	0.1341	0.1046–0.1540	0.1205	0.1745
M33	104	0.0876	0.0717–0.1043	0.0847	0.1460
M34	104	0.0819	0.0435–0.1008	0.0731	0.1629
M41	104	0.1384	0.1031–0.1534	0.1273	0.1796
M42	104	0.1456	0.1312–0.1573	0.1385	0.1796
M43	104	0.0895	0.0772–0.1307	0.0983	0.1742
M44	104	0.0894	0.0800–0.1102	0.0945	0.1468

Table S1: Within-specimen variability of Markov transition parameters across 100 stochastic fits for MRD (flow).

Transition	N	Median within-specimen SD	IQR of within-specimen SD	Mean within-specimen SD	Maximum within-specimen SD
M11	101	0.0788	0.0227–0.1101	0.0692	0.1482
M12	101	0.1102	0.0171–0.1352	0.0840	0.1710
M13	101	0.0144	0.0005–0.1287	0.0586	0.1631
M14	101	0.0129	0.0009–0.0747	0.0373	0.1562
M21	101	0.0936	0.0205–0.1315	0.0800	0.1655
M22	101	0.1135	0.0591–0.1323	0.0986	0.1682
M23	101	0.0252	0.0019–0.0992	0.0506	0.1568
M24	101	0.0098	0.0014–0.0748	0.0391	0.1595
M31	101	0.1399	0.0825–0.1493	0.1106	0.1720
M32	101	0.1319	0.1018–0.1519	0.1173	0.1745
M33	101	0.0873	0.0713–0.1047	0.0846	0.1460
M34	101	0.0809	0.0227–0.1013	0.0708	0.1629
M41	101	0.1341	0.1007–0.1528	0.1257	0.1796
M42	101	0.1447	0.1305–0.1558	0.1379	0.1739
M43	101	0.0934	0.0775–0.1336	0.1004	0.1742
M44	101	0.0912	0.0802–0.1124	0.0952	0.1468

Table S2: Within-specimen variability of Markov transition parameters across 100 stochastic fits for BCR::ABL1.

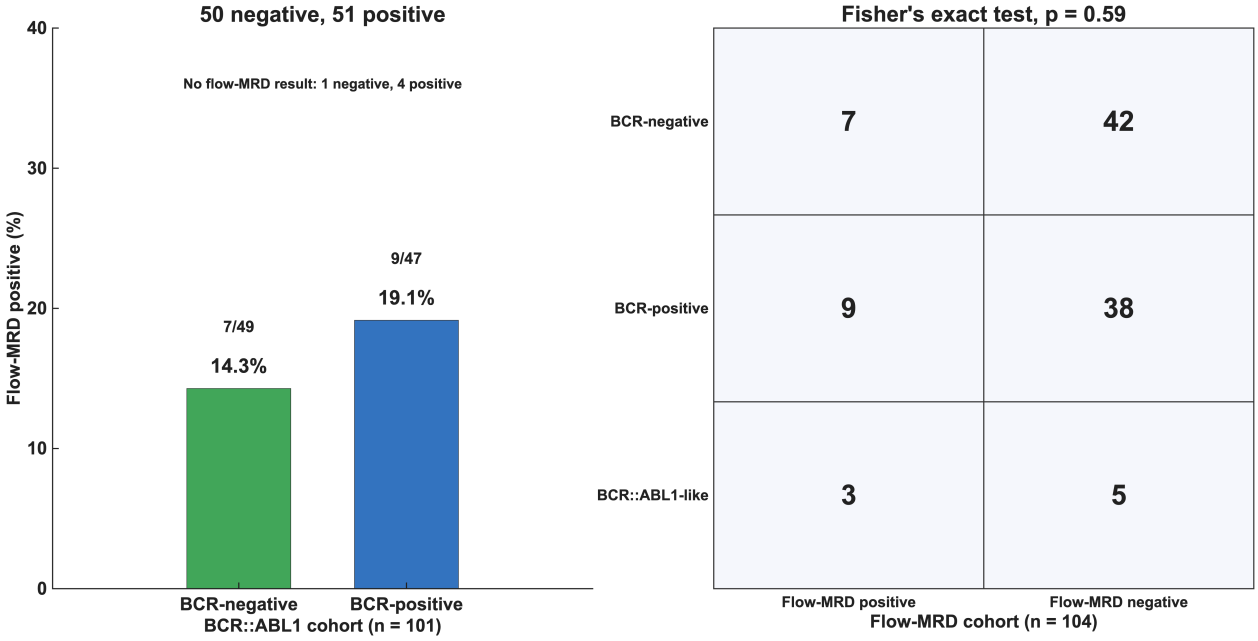


Figure S2: Counts of specimens by BCR::ABL1 fusion status and post-induction MRD status. Fisher's exact test did not detect a statistically significant association between the outcomes in this cohort.

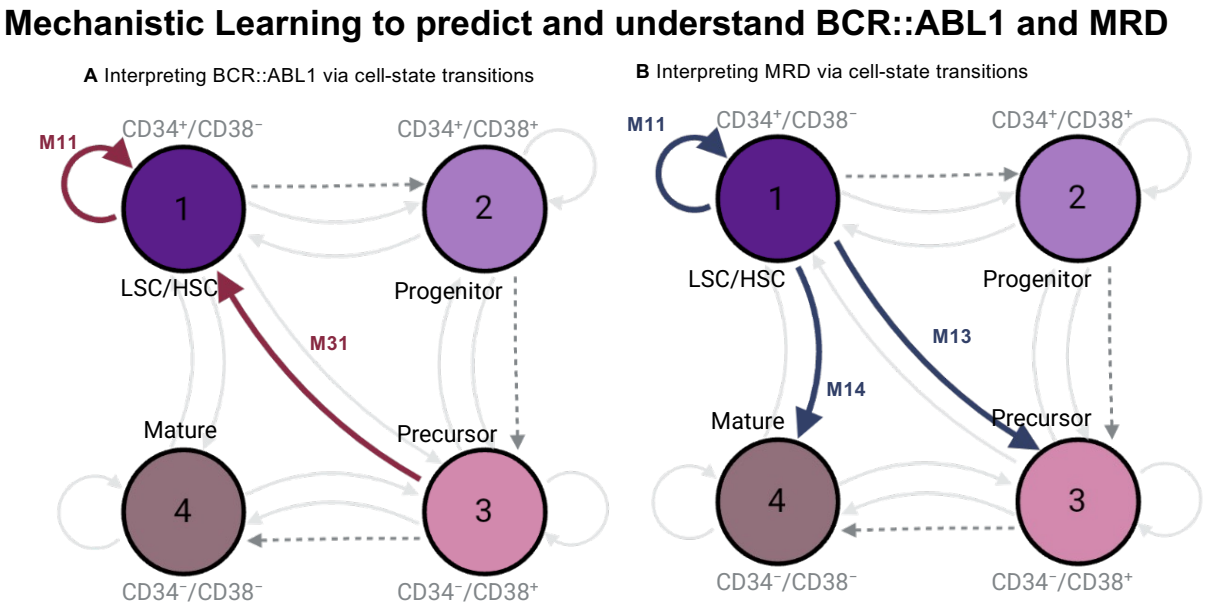


Figure S3: Four CD34/CD38-defined phenotypic states and the Markov model-inferred transitions most frequently selected as predictive features for BCR::ABL1 fusion status and post-induction MRD.